

Purpose and scope

This document provides a technical review of the recent structural changes applied to the IDTA Submodel Template EPD Type III (Environmental Product Declaration). It is intended as a basis for decision-making and technical review by the working group. It focuses on the transition from hard-coded indicator structures to a generic, flexible IEC 61360-based structure, the normalization of identifiers, and the handling of characterization units.

Executive summary

The EPD Type III Submodel Template was fundamentally refactored to use a dynamic, template-based approach for LCIA results and resources instead of statically predefined indicator groups. Key improvements include dynamic `idShort` generation in instances for better tooling support (while keeping templates compliant with AASd-120), a refactored and cleaner `characterizationUnit` hierarchy, and successful validation against a real-world WAGO PEP. Several potential redundancies and missing fields were identified during the instantiation process, which are documented here for working group discussion.

Overview of implemented changes

Change ID	Affected path	Previous design	Current design	Normative basis	Review question
CHG-001	<code>lciaResults/*</code>	Predefined static SubmodelElementCollections for each indicator group	Single generic template collection in SMT. Dynamic <code>idShort</code> (e.g. <code>GWP_total</code>) generated during instantiation.	AAS Part 1 V3.0 (AASd-120 restricts <code>idShort</code> in SML)	Do reviewers agree that <code>idShort</code> values for LCIA list entries shall only be assigned during instantiation and not fixed in the generic template?
CHG-002	<code>*/lifeCyclePhaseValues/*</code>	Elements named "Unknown" in UI	Dynamic <code>idShort</code> (e.g., <code>A1_A3</code>) derived from <code>lifeCyclePhase</code> value.	AAS Part 1 V3.0	Should <code>Total</code> remain an entry in <code>lifeCyclePhaseValues</code> , or should it be represented as a separate <code>totalValue</code> property?
CHG-003	<code>*/characterizationUnit</code>	Flat or mixed structure	Nested structure: <code>unit</code> , <code>unitId</code> , optional <code>baseUnit</code> (<code>unit</code> , <code>unitId</code>).	IEC 61360	Do reviewers agree that the full LCIA characterization unit remains authoritative, while the ECLASS base unit is optional supplementary information?
CHG-004	List semantics	Missing list semantics	<code>typeValueListElement</code> set to SMC. <code>semanticIdListElement</code> added in instance.	AAS Part 1 V3.0	Are the applied list element semantics sufficient for standardizing the EPD structure?

Detailed change descriptions

1. idShort for LCIA results

- Previous representation:** In older templates, or strictly standard V3 lists, elements without `idShort` are displayed as `Unknown` or generically (e.g., `SubmodelElementCollection`) in the AASX Package Explorer.
- New representation:** Direct list elements of `lciaResults` in a concrete instance now receive an `idShort` derived from the `indicatorCode` (e.g., `indicatorCode: GWP-total` → `idShort: GWP_total`).
- Normalization rules:** Spaces, dashes, and slashes are replaced by underscores. Non-alphanumeric characters are removed. Consecutive underscores are collapsed.
- Conflict handling:** If an `idShort` conflict occurs (e.g., duplicate indicators), a suffix `_2` is appended.
- Template vs. Instance:** In the generic template, the template entry has no `idShort` to comply with AAS V3 rule AASd-120 (direct children of a `SubmodelElementList` must not have an `idShort`). In the concrete instance, however, it is dynamically assigned to drastically improve UX in viewers like the AASX Package Explorer.
- AAS Metamodel Impact:** AAS V3.0 restricts `idShort` usage in lists. Tools might flag this as a warning, but the UX benefit for human readability is significant.

2. idShort for Life Cycle Phases

- Analogous to LCIA results, `idShort` values are assigned to direct list elements of `lifeCyclePhaseValues` in the instantiated model.
- Examples:** `A1-A3` → `A1_A3`, `B1-B7` → `B1_B7`, `C1-C4` → `C1_C4`, `D` → `D`, `Total` → `Total`.
- The actual normative value in the `lifeCyclePhase` property remains untouched (e.g., `A1-A3`). The `idShort` is strictly for unique identification within the model and better tool representation.

3. Modeling of Characterization Units

- **Compact Structure:** The structure was revised to cleanly separate the LCIA impact unit from the physical unit.

```
```text
characterizationUnit
├── unit (e.g., "kg CO2 eq.")
├── unitId (e.g., "")
└── baseUnit [optional]
 ├── unit (e.g., "kg")
 └── unitId (e.g., "0173-1#05-AAA731#005")
...
```
```

- `unit` designates the full LCIA characterization unit.
- `baseUnit` designates the physical base unit (e.g., from ECLASS). The ECLASS unit *does not* replace the LCIA unit; it is supplementary.
- `baseUnit` is strictly optional.
- Having identical child names (`unit`, `unitId`) is valid as they belong to different parent collections.
- Non-verified ECLASS units are not guessed. Custom LCIA `unitId` URIs are used as preliminary working identifiers until official IDTA/ECLASS IDs are confirmed.

4. Changes to Lists and Semantics

- **semanticIdListElement:** In the instance generation, a `semanticIdListElement` is explicitly injected into `lifeCyclePhaseValues` to formally define the semantics of the list items.
- **typeValueListElement:** Defined as `SubmodelElementCollection` for all generic lists to restrict list contents.
- **Cardinalities:** Kept flexible (`ZeroToMany`) for template definitions to allow arbitrary numbers of indicators in instances.
- **AAS Version:** Maintained at AAS V3.0. List semantics conform strictly to V3 specifications.

Validation with the WAGO PEP

The dynamic model was tested against a real-world Product Environmental Profile (PEP): [WAGO-00001-V01.01-EN.pdf](#).

- **Mapped Metadata:**
 - Registration Number: `WAGO-00001-V01.01-EN` (Version: `V01.01-EN`)
 - Dates: Issue Date (`02-2026`), Valid Until (`02-2031`), Validity Period (`5 years`)
 - Verifier Accreditation: `VH44`
 - PCR Reference: `PCR-4-ed4-EN-2021 09 06`
 - Product Data: `221 Series / Splicing Connector with Levers`, Reference `221-422`, Flow `1.958 g`, Service Life `30 years`.
- **Mapped Indicators:** 20 LCIA indicators, 10 Resource Use indicators, 3 Waste categories, 4 Output flows.
- **Life Cycle Phases:** Successfully mapped values for phases `A1-A3` (aggregated), `A4`, `A5`, `B1-B7` (aggregated), `C1-C4` (aggregated), and `D`.
- **Value Handling:** Negative values (e.g., carbon sequestration) and scientific notation values were correctly processed and retained as `xs:decimal`.
- **Conclusion:** The instance (`WAGO-00001-V01.01-EN_AAS31.aasx`) serves as a practical proof of concept that the restructured, generic SMT can successfully and robustly handle complex, real-world EPD data.

Potential redundancies

| Affected Elements (AAS Path) | Overlap Explanation | Recommendation | Status |
|---|--|--|-------------------|
| <code>epdId</code> vs <code>registrationId</code> | Both fields frequently store the exact same PEP/EPD registration number. | Merge / Clarify definitions | Proposed |
| Declared Unit Modeling | Declared unit is often implicitly defined in metadata and explicitly in <code>referenceUnitForCalculation</code> . | Clarify primary source of truth | Open for decision |
| <code>kgPerDeclaredUnit</code> vs <code>massPerDeclaredUnit</code> | Both describe the mass of the declared unit. | Merge / Rename to <code>massPerDeclaredUnit</code> | Proposed |
| <code>productServiceLifeYears</code> vs <code>referenceServiceLife</code> | Both specify the service life. Often identical data. | Merge to <code>referenceServiceLife</code> | Proposed |
| Reference Flow vs Declared Unit | Reference flow and declared unit are often synonymous in PEPs but modeled in parallel. | Retain, but add clear guidance | Open for decision |

Information not yet represented

The following information fields from the WAGO PEP currently lack a standardized, dedicated property in the model:

- Structured functional unit (currently mapped as plain text)
- Separate product mass and packaging mass (currently only total/reference flow)

- Detailed material composition
- Product families and covered variants
- Indicator- and phase-specific extrapolation factors
- Verifier accreditation number (`VH44`) - mapped, but needs formal review for correct placement
- Structured information on LCA software, background database, and LCIA method
- Status of individual modules (e.g., distinguishing between `declared` , `not applicable` , `excluded` , `aggregated`)

Note: These are documented as review findings; no template changes were made for these items yet.

Backward compatibility and migration impact

- **Breaking Changes:** Moving from static indicator collections to a single generic list (`lciaResults`) is a breaking structural change. Existing parsers expecting hard-coded paths (e.g., `lciaResults/GWP-total`) will fail.
- **Migration Path:** Software extracting EPD data must be updated to iterate over `lciaResults` lists and dynamically check the `indicatorCode` property instead of relying on the structural `idShort` .
- **AAS Version:** Retained V3.0.

Validation results

- AASX file can be opened.
- File can be saved and reopened without corruption.
- `idShort` values in the instance are successfully retained.
- LCIA indicators are no longer displayed as `Unknown` in the Package Explorer.
- Life cycle phases are no longer displayed as `Unknown` .
- Numeric values, signs (including negatives), and units are fully retained.
- `semanticIdListElement` and list element semantics are structurally consistent.
- Template (generic) and Instance (specific) architectures are clearly separated.
- Known tool warnings (e.g., `idShort` in Lists) are distinguished from actual schema failures.
- AAS version explicitly maintained at 3.0.

No formal conformance to XML schemas is asserted here, as AASd-120 warnings are expected when `idShort` is used in lists. This was a conscious decision to improve UX.

Open decisions for the working group

1. **List IDShorts:** Do we formally accept the addition of `idShort` to List elements in *instances* to improve UI representation, despite AASd-120?
2. **Characterization Units:** Is the `baseUnit` approach sufficient, and do we agree to use preliminary URIs for LCIA units until ECLASS provides them?
3. **Total Values:** Should `Total` be handled as just another phase in `lifeCyclePhaseValues` , or separated as a distinct property to prevent aggregation errors?
4. **Redundancies:** Which of the identified potential redundancies (`epdId` vs `registrationId` , etc.) should be merged in the next template release?

Reviewer comment table

(Please see the accompanying `EPD_Type_III_Submodel_Review_Comments.xlsx` for submitting feedback.)

| Comment ID | Reviewer | Section / Change ID | Comment | Proposed resolution | Decision | Status |
|------------|----------|---------------------|---------|---------------------|----------|--------|
| C-001 | | CHG-001 | | | | |
| C-002 | | CHG-003 | | | | |

Change log

- **2026-07-15:** Initial draft of structural changes for review. Validated against WAGO PEP.

Appendix: before/after structures

Before (Static, hardcoded):

```
lciaResults (SubmodelElementCollection)
├─ GWP_total (SubmodelElementCollection)
|   └─ indicatorCode: "GWP-total"
|   └─ lifeCyclePhaseValues
|       └─ (Unknown SMC) -> lifeCyclePhase: "A1-A3", value: 1.2
└─ ODP (SubmodelElementCollection)
```

└ ...

After (Generic List with dynamic instantiation):

```
lciResults (SubmodelElementList)
└─ GWP_total (SubmodelElementCollection) [In Instance Only]
  │ └─ indicatorCode: "GWP-total"
  │ └─ characterizationUnit
  │   │ └─ unit: "kg CO2 eq."
  │   │ └─ baseUnit (optional)
  │ └─ lifeCyclePhaseValues (SubmodelElementList)
  │   │ └─ A1_A3 (SubmodelElementCollection) [In Instance Only]
  │   │   │ └─ lifeCyclePhase: "A1-A3"
  │   │   │ └─ value: 1.2
```